ADITI AWASTHI

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Summary

PhD student researching architectural design for Fault-Tolerant Quantum Computers (FTQC), with a focus on quantum resource estimation, error correction, and circuit optimization and compilation. Prior industry experience includes two years at Goldman Sachs building low-latency data pipelines for quantitative trading.

Education

The University of Texas at Austin

PhD in Electrical and Computer Engineering — Aug 2024 - May 2028 (Expected)
Research Advisor: Dr. Jonathan Baker
Research Focus: Quantum Computing Architecture, Compilation and Error Correction

MS in Computer Science — Aug 2022 - May 2024
GPA: 3.82/4.00
Relevant coursework: Quantum Information Science I and II, Quantum Computing Systems: Software and Architecture, Quantum Error Correction, Quantum Complexity Theory, Unconventional Computation, Algorithms: Techniques and Theory, Linear Algebra, Cryptography, Computer Architecture

B.M.S. College of Engineering, Bengaluru

BE in Computer Science and Engineering — Aug 2016 - Aug 2020
GPA: 9.83/10.00

Research Experience

The University of Texas at Austin

- **Dr. Jonathan Baker's Lab** — Aug 2024 - Present
  Conducting research in quantum computing architecture and error correction geared towards developing robust quantum circuit compilation and optimization techniques. Notable projects include:
  - **Quantum Resource Estimation for FTQC**
    Modeled magic-state factory allocation for fault-tolerant quantum circuits, and characterized spacetime trade-offs across distillation, cultivation, and R_z-synthesis regimes under realistic error models.
  - **Pipelining and Speculation for FTQC**
    Designed and analyzed speculative execution and load-balancing strategies for quantum circuits.
  - **Trade-offs involved in decoding algorithms for surface codes**
    Analyzed Sparse Blossom decoder performance under limited parallelization and partitioning constraints, identifying optimization strategies for latency vs. accuracy trade-offs.
- **Dr. David Soloveichik's Lab** — Jun 2023 - May 2024
  Conducted research in unconventional computing, specifically computing concentrations of domain-level DNA designs for molecular programming systems.

B.M.S. College of Engineering, Bengaluru

Dr. Kayarvizhy N's Lab Aug 2019 - Aug 2020

Refined the solution to the de-fencing issue in image enhancement by using frequency-domain analysis, signal processing and connected component labelling for fence mask refinement.

Research Publications

1. **A. Awasthi**, G. S. Ravi, and J. M. Baker, *Stalls and spequlation: Pipelined execution for fault tolerant quantum computation*, 2026. arXiv: 2606.19593 [quant-ph]. [Online]. Available: https://arxiv.org/abs/2606.19593 (submitted to QCE 2026)
2. **A. Awasthi**, S. Sethi, S. Khan, G. S. Ravi, and J. M. Baker, *Price and payoff: Non-determinism in fault tolerant quantum computation*, 2026. arXiv: 2605.07983 [quant-ph]. [Online]. Available: https://arxiv.org/abs/2605.07983 (submitted to QCE 2026)
3. S. Sethi, S. Khan, **A. Awasthi**, A. Anand, and J. M. Baker, *Injeqt: Improved magic-state injection protocol for fault-tolerant quantum extractor architectures*, 2026. arXiv: 2604.25094 [quant-ph]. [Online]. Available: https://arxiv.org/abs/2604.25094 (submitted to QCE 2026)
4. **A. Awasthi**, D. Bhat, M. Oak, and N. Kayarvizhy, "Effectiveness of connected components labelling approach in noise reduction for image de-fencing," in *Proceedings of International Conference on Communication and Computational Technologies*, Springer, 2021, pp. 635–646. DOI: 10.1007/978-981-16-3246-4_49
5. P. Bambroo and **A. Awasthi**, "Legaldb: Long distilbert for legal document classification," in *2021 International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)*, 2021, pp. 1–4. DOI: 10.1109/ICAECT49130.2021.9392558

Professional Experience

Goldman Sachs, Bengaluru, India

Engineering Analyst, Synthetic Product Group (SPG) Sep 2020 - Jun 2022

- Built high-performance data pipelines for quantitative trading clients, optimizing for scalability and low-latency processing (synthetic/swap reporting)

Engineering Intern, Synthetic Product Group (SPG) Jan 2020 - Jun 2020

- Enriched the synthetic reporting service offering for clients trading on the dark pool.
- Developed an automated regression testing tool for swap booking to improve software quality.

Engineering Intern, Ledgers Technology Jun 2019 - Aug 2019

Enhanced the scale of the firmwide-ledger exception identification systems, to enable efficient processing and flagging of transactions.

Indian Institute of Technology, Kanpur

Software Development Intern, Centre for Cyber Security Jun 2018 - Jul 2018

Mentor: Dr. Manindra Agrawal

Developed a Threat Intelligence System to detect and analyze cyber attacks. This included building parsing tools that operate on Apache logs, and creating analytical dashboards using the MERN stack.

Other Academic Experience

The University of Texas at Austin

Graduate Teaching Assistant

- **Unconventional Computation** *ECE381V/CS395T* — Spring 2024, Spring 2025, Spring 2026
  Assisted instruction across different computation paradigms including analog, chemical, reversible, and quantum computing.
- **Algorithms: Techniques and Theory** *CS388G* — Spring 2023
  Assisted instruction in advanced algorithm design and analysis, covering maximum flow, randomized algorithms, NP-completeness, and approximation algorithms.
- **Software Engineering and Design Lab** *ECE461L* — Fall 2025
  Supported students in hands-on learning of full-stack web development, software design patterns, and collaborative team projects using tools like Python, React.js, MongoDB, and Git.

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Undergrad Projects

- *Automated Answer Script Evaluation*
  Handwritten text recognition tool for automated exam grading against a rubric.
- *CNC (Computer Numerically Controlled) Drawing Machine*
  Arduino-powered device translating digital vector images into precise paper sketches.
- *Twitter Sentiment Mining and Analysis*
  R-based tool to visualize and analyze public opinion on trending topics.
- *Institutional Course Elective Portal* (Pan-college project undertaken by CS Developer Team)
  Web application to streamline cross-listed course selection for ∼5000 students.

Awards

Charles M. Simmons Endowed Presidential Fellowship in Engineering — 2026 – 2027
Awarded by the Cockrell School of Engineering (UT Austin).

BMSCE Alumni Network merit scholarship (6-time recipient) — 2016 - 2020
Awarded for outstanding academic achievements in each semester.

Extracurriculars

BMSCE IEEE Student Branch, Vice-Chairperson — 2018 - 2019
Enabled students (across several majors) to up-skill themselves, by curating seminars and workshops.

IEEE Python Special Interest Group, Instructor — Aug 2017 - Nov 2017
Planned the 'Programming with Python' course syllabus and instructed a class of 50+ undergrad students.

PhaseShift Technical Fest, Department Coordinator — Jun 2018 - Sep 2018
Organized departmental events for BMSCE's annual technical fest.

Technical Skills

Quantum Computing: IBM Qiskit, Stim, Sinter

Programming: Python, Java, C++, R, SQL, Linux Shell

Tools & Frameworks: MongoDB, Hadoop, Apache Spark, React.js, Git